YIDING MAO

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EDUCATION

University of Nottingham Ningbo China (2+2) Ningbo, China
Bachelor of Science in Statistics Sep 2023 to Present
University of Nottingham Nottingham, UK
Bachelor of Science in Statistics Sep 2025 to Present
Cumulative GPA: 3.98/4.00 (top 3%)
Relevant Courses: *Algebra, Complex Analysis, Optimization, Linear Models, Statistical Modelling with Machine Learning, Stochastic Models, Markov Chains, Scientific Computation*

RESEARCH AND PROJECT EXPERIENCE

A Deep Reinforcement Learning-Based Strategy for Instant Retail Supply Chain Interruption (repo) Ningbo, China
Supervised by Dr. Ruijing Wu *Dec 2025 to Present*

- Formulated instant-retail disruption response as a Markov decision process (MDP), modeling supplier-capacity and transport disruptions as Markov-modulated states, encoding buffer level, expediting and allocation priority as 18 discrete actions with physically motivated masking, and defining an inventory position that includes capacity and transport backlogs to prevent duplicate ordering; implemented the formulation as a simulator of a two-echelon network with 10 forward warehouses and 5 products.
- Developed a reproducible training and evaluation pipeline for 25 agents (5 DRL algorithms × 5 seeds) over 2.25M simulated days, incorporating common random numbers, independent random-number streams for demand, disruptions and lead times, and risk-augmented initial states to improve coverage of rare, severe disruption states.
- Designed an evaluation protocol addressing seed sensitivity and pseudo-replication documented in DRL research, with the training seed as the statistical unit and paired confidence intervals; established that no DRL agent outperformed a base (s, S) policy and attributed this result to behavioral and recovery mechanisms, forming the central contribution of a completed manuscript.
- Identified structural limitations of the benchmark formulation and developed a revised framework: a lost-sales MDP in which disruptions also delay in-transit inventory, a masked autoregressive PPO policy generating joint integer purchase and shipment decisions, a proof that the policy generates exactly the feasible action set, and an exact bounded-cost reformulation preserving policy rankings.

World Model for Autonomous Systems Ningbo, China
Supervised by Dr. Huan Jin in AIOP Research Group *Jun 2026 to Present*

- Curated a 20,000+ frame dataset from 80 GB of self-collected real-world navigation video, labeled with spatio-temporal poses derived from relative positions and trajectory yaw angles; trained NoMaD (Goal Masked Diffusion Policy) on this dataset with dynamic loss scaling to balance the action and distance objectives, using trajectory-direction cosine similarity as the early-stopping criterion.
- Conducted a controlled ablation over training duration (5 vs. 10 epochs, 15 checkpoints); extended training raised trajectory-direction cosine similarity from approximately 0.94 to above 0.97 and reduced both goal-conditioned distance and undirected action losses, whereas matched-epoch checkpoints differed across independent initializations, indicating seed-level variance that single-run comparisons cannot capture.
- To enable model-based evaluation of candidate actions, reproduced the Conditional Diffusion Transformer (CDiT)-based Navigation World Model for autoregressive prediction of future visual observations from past observations and actions.
- Integrated the two models by employing the world model to score and rank NoMaD-sampled candidate trajectories offline; identified that a subset of predicted frames, although visually plausible, deviated from ground-truth observations, indicating that visual plausibility alone is an insufficient criterion for validating world-model-based planning.

Clinical Mechanical Ventilation Modeling Remote
Supervised by Dr. Mainul Haque *Jul 2025 to Aug 2025*

- Built an ODE model of mechanical ventilation capturing airway pressure, intrinsic PEEP and arterial blood-gas dynamics, and estimated its parameters by numerical optimization in Python (NumPy) and MATLAB.
- Simulated airway pressure and gas-exchange efficiency across PEEP settings and reproduced published findings on the effect of PEEP in chronic airflow obstruction, validating the model against the literature.

PROFESSIONAL EXPERIENCE

Zhejiang Humanoid Robot Co., Ltd. Ningbo, China
Platform Software Intern *Jun 2026 to Sep 2026*

- Developed ROS 2 nodes to read robot telemetry and traced 10+ telemetry and event streams through the company's data platform (Apache Pulsar, IoTDB, Doris and MySQL on Kubernetes), from real-time ingestion to time-series storage and analytical querying.
- Built data-quality validation workflows that reconciled timestamps and cross-source identifiers and flagged missing, duplicated and inconsistent records, reaching 95%+ cross-source record matching for downstream analytics and debugging.
- Worked on the tour-guide, data-collection and teleoperation apps on physical robots; diagnosed a relocalization-status display failure by tracing from the UI through ROS communication to robot-side topics, isolated the fault to the app code, and verified the fix on-robot after release.
- Extended an LLM-based code-review system built on Alibaba Open-Code-Review with diff-aware context retrieval and a code knowledge graph, increasing true-positive defect detection by 7.7%; proposed a no-plan mode for rule-matched tasks that cut review time to ~1/7 and cost by ~50% on an internal sample.

BOE Technology Group Co. Beijing, China
Data Analyst Intern *Jun 2025 to Aug 2025*

- Built an employee-attrition analytics framework on 50,000+ records from 2022–2025, integrating personnel, factory, operational and external benchmark data to support workforce-risk planning.
- Cleaned and standardized multi-source records, handled missing and anomalous values, and engineered 20+ employee- and department-level features, including overtime intensity, compensation deviation, tenure and historical turnover.
- Used linear regression for department-level turnover forecasting and compared logistic regression with decision-tree classifiers for individual attrition prediction; the best classifier reached 0.84 AUC and 78% recall.
- Identified the key predictors of attrition and translated them into Power BI dashboards and retention recommendations for departmental stakeholders.

Donghai Insurance Ningbo, China
Data Analyst Intern *Jul 2024 to Aug 2024*

- Built a Python-based data preprocessing workflow to clean and structure 20,000+ policy and claims records, constructing policyholder- and policy-level features for pricing analysis.
- Developed frequency and severity models to decompose expected loss — Poisson regression with a log-exposure offset to model annualized claim frequency, and a log-normal severity model fitted on log-transformed claim amounts over the claims-incurred subset — using 15+ policyholder and policy features; decision trees were fitted as a benchmark and to surface candidate interactions.
- Combined predicted frequency and severity to estimate expected loss and compare loss ratios across risk segments; validated model stability with 5-fold cross-validation and used feature-importance analysis to identify the dominant risk factors, which informed risk-tier and premium-rating recommendations.

SKILLS

Programming: Python, R, SQL, MATLAB
ML & Data Science: PyTorch, scikit-learn, pandas, NumPy
Data Systems: Apache Pulsar, IoTDB, Doris, MySQL, Kubernetes
Tools: Git, Linux, ROS 2, Power BI, Tableau